MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBOTA07 - PRINCIPLES OF AGRONOMY AND HORTICULTURE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Part A – Very Short Answer (2 marks each)
(Award 1 mark per point)
Answer any 10 questions
(Minimum one point for full mark)

1. Explain horticulture.
• Branch of agriculture
•Cultivation of fruits, vegetables, flowers, ornamentals

2. Define secondary tillage.
• Operation after primary tillage
• Soil refinement and seedbed preparation

3. Define layering.
• Vegetative propagation method
• Roots develop on stem attached to parent plant

4. Write the recommended age of solanaceous seedlings for transplanting.
• 25–35 days or
• 4–6 weeks

5. Define nutrient deficiency in plants.
• Lack of essential nutrients
• Leads to deficiency symptoms

6. What is meant by shade house nursery?
• Nursery covered with shade net
• Controls light and temperature

7. Name any two components of INM.
• Chemical fertilizers
• Organic manures / biofertilizers

8. What is urea?
• Nitrogenous fertilizer
• Contains 46% nitrogen

9. Abiu belongs to which family?
• Sapotaceae

10. What is meant by indigenous agricultural practices?
• Traditional farming methods
• Based on local knowledge

11. Explain the objectives of White Revolution.
• Increase milk production
• Improve dairy sector

12. What is meant by laterite soil?
• Rich in iron and aluminium
• Formed in high rainfall regions

PART B – Short Essay (5 marks each)

Answer any 8 questions

(Minimum 4 points with brief explanation for full mark)

13. Discuss the use of shrubs and trees in garden planning and landscaping.
• Provide structure – form the framework and layout of the garden
• Give shade – reduce temperature and create comfortable spaces
• Improve aesthetics – enhance beauty with flowers, foliage, and form
• Act as windbreak – reduce wind speed and protect plants
• Provide privacy – act as screens or hedges
• Support biodiversity – attract birds, insects, and beneficial organisms

or any other relevant points

14. Explain how biological agents help in sustainable pest management.
• Use of predators – insects like ladybird beetles feed on pests
• Use of parasitoids – organisms like *Trichogramma* lay eggs in pests
• Use of microbes – bacteria and fungi kill harmful insects
• Eco-friendly – no harmful chemical residues
• Reduces chemical pesticides – lowers environmental pollution

or any other relevant examples

15. Describe the procedure for preparing and planting cuttings.
• Select healthy plant – ensures disease-free propagation
• Prepare cuttings – stem pieces with nodes are used
• Remove lower leaves – reduces water loss
• Apply rooting hormone – promotes root formation
• Plant in medium – sand or soil mix supports rooting
• Maintain moisture – essential for survival and growth

16. Write a note on the modern developments in Indian agriculture.
• Green Revolution – introduction of HYV seeds increased yield
• Biotechnology – development of improved and resistant varieties
• Precision farming – efficient use of inputs like water and fertilizers
• Advanced irrigation – drip and sprinkler systems conserve water
• Digital agriculture – use of AI, sensors, and mobile apps

17. Explain the process and benefits of vermicomposting.
• Use of earthworms – species like *Eisenia fetida* decompose waste
• Decomposition of organic waste – converts biodegradable waste into compost
• Produces nutrient-rich compost – high in NPK and micronutrients
• Improves soil fertility – enhances soil structure and microbes
• Eco-friendly – reduces waste and pollution

or any other relevant examples

18. Explain the advantages and disadvantages of budding.
• True-to-type plants – maintains genetic characters
• Early bearing – plants mature faster
• Economical – requires less plant material
• Requires skill – proper technique is essential
• Seasonal limitation – only done in suitable seasons
• Limited applicability – not suitable for all crops

19. List and describe the sequential steps involved in hybridization.
• Selection of parents – choose desirable traits
• Emasculation – removal of anthers to prevent self-pollination
• Bagging – cover flowers to avoid contamination
• Pollination – transfer pollen manually
• Tagging – label with details of cross
• Seed collection – harvest and evaluate hybrid seeds

20. Describe the current trends in agricultural policies and practices in India.
• Focus on sustainability – promoting eco-friendly farming
• Promotion of organic farming – reducing chemical inputs
• Subsidies and MSP – ensuring farmer income stability
• Crop insurance – protection against crop failure
• Technology adoption – digital tools and smart farming

21. Explain the role of agronomy in sustainable crop production.
• Efficient resource use – proper use of water, nutrients, and land
• Maintains soil fertility – through crop rotation and organic inputs
• Crop rotation – prevents pest buildup and improves soil health
• Water management – efficient irrigation practices
• Pest control – integrated pest management methods

22. Discuss the factors affecting soil fertility.
• Soil texture – affects water and nutrient holding capacity
• Soil structure – influences aeration and root growth
• Organic matter – improves nutrient availability
• pH – controls nutrient solubility
• Microorganisms – help in nutrient cycling
• Climate – affects weathering and biological activity

PART C – Essay (10 marks)

Answer any 1 question

23. Describe the different types of grafting and their applications in horticulture.

• Approach grafting – scion and stock remain attached until union forms
• Cleft grafting – scion inserted into split stock
• Spice Grafting–It is a method of grafting by cutting the ends of the scion and stock completely across and obliquely
• Whip and tongue grafting – interlocking cuts ensure strong union
• Side grafting – scion inserted into side of rootstock
• Propagation of fruit crops – widely used in mango, citrus, apple
• Combining desirable traits – disease resistance and high yield
• Disease resistance – resistant rootstock improves plant health
• Early bearing – grafted plants produce fruits faster
• Uniform plants – ensures consistency in orchard

Any of the above 4 types with explanations

24. Give an essay on disease and pest management in Mango.

Important pests of mango are hoppers, stem borers, shoot midges, leaf-feeding insects, fruit flies and mealy bugs. The common diseases are powdery mildew, anthracnose and dieback.

the following steps can be practiced to protect the orchard and trees:

- Keep the orchard and tree surroundings clean by proper sanitary measures.
- Lime brushing on tree trunks will help to control some pests.
- Cuts and wounds on plant parts should be pasted with coal tar, used engine oil or Bordeaux paste.

- Smoking in low intensities in the orchard during flowering season will reduce the number of pests including hoppers.
- Spraying Pseudomonas fluorescens (10 g/litre) on trees before flowering and at flushing times will improve the general health, flowering and resistance of the trees.
- To control fruit flies, collect and destroy attacked fruits that rot and drop down.
- Banana fruit traps can be used by mixing banana fruits (Palayanthodan) and jaggery (3–5%) poisoned with 2 ml malathion, and keeping them in hanging containers or using for swabbing tree trunks at weekly intervals from flushing time onwards.
- Set up methyl euginol trap @ 10 nos/ha.
- Spathiphyllum (Peace Lily) plants can be planted as trap crop and the attracted insects can be destroyed manually.
- To control stem borer, apply paste made of crude carbolic acid (130 ml), soft soap (1 kg) and hot water (3.7 litres) to holes in the bark and plug the holes.
- Against sucking insects such as mealy bugs, thrips and mites, spray neem oil (0.5 to 1%) – soap emulsion.
- Sticky bands or stem traps with mud slurry, jack latex, vaseline, grease, coal tar or gel will prevent the upward movement of pests from soil.
- Red ants may become serious sometimes. If colonies are maintained at a moderate level, they help guard the trees from common insect pests. If severe, control can be attempted by introducing predatory ant, Plagiolepis (Chonal ants), and by mechanical means such as breaking ant nets, sprinkling wood ash or common salt pellets, and trapping ants.
- To control dieback, cut the affected twigs below the infected region and spray 1% Bordeaux mixture.